



THE ROLE OF DIGITAL TECHNOLOGIES IN OPTIMIZING SUSTAINABLE TRANSPORTATION IN MODERN SUPPLY CHAIN NETWORKS



Abstract

- This research explores how digital technologies, such as IoT, AI, ML, and route optimisation software, can be employed to encourage sustainable transportation logistics
- The study was structured with positivist philosophy, deductive approach and descriptive design.
- Quantitative data that will be gathered through an online survey that will be sent to supply chain professionals.
- The results showed that 50.4 percent of the respondents reported AI and blockchain as efficiency drivers.
- 67.4 percent acknowledged that digital tools help to cut emissions and reinforce environmental performance.
- Citation of organisational level of obstacles aid in limiting adoption such as skills or costs

Introduction

- Sustainable transport is a pillar of the UN 2030 Agenda since transport accounts for more than 8 billion metric tons of carbon dioxide emissions every year, ranking second in terms of global greenhouse gas emissions.
- Along with large international logistics companies being at the forefront, the SMEs encounter obstacles in terms of exorbitant costs, the unavailability of skilled staff, insufficient digitalisation, and regulatory disparities.
- Businesses and policymakers are turning their heads to emerging technologies as a means of reducing these environmental issues whilst sustaining operational performance (Ojadi et al. 2024).
- The COVID-19 pandemic demonstrated the inflexibility of traditional supply chain models and the necessity of flexible, technology-based solutions.
- Sustainable supply chain management (SSCM) is the management of supply chain activities, resources, information, and money to the greatest social benefit (Kazancoglu et al. 2023).

Method

- Study was done based on positivism, which guaranteed objectivity and collecting quantifiable data (Taherdoost, 2022)
- The deductive method challenged the available theories of the importance of technology in sustainability.

Independent Samples Test										
Levene's Test for Equality of Variances						t-test for Equality of Means				
On a scale of 1-5, how satisfied are you with the performance of digital tools currently used in your transport network?		F	Sig.	t	df	Sig. (2-tailed)	Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
									Lower	Upper
	Equal variances assumed	.860	.356	1.224	127	.223	.201	.164	-.124	.525
	Equal variances not assumed			1.208	99.998	.230	.201	.166	-.129	.530

ONEWAY Tech_Impact_Efficiency_encoded BY Org_Size_encoded
/MISSING ANALYSIS
/POSTHOC=Tukey ALPHA(.05).

T-Test				
Group Statistics				
	What is your age group?	N	Mean	Std. Deviation
On a scale of 1-5, how satisfied are you with the performance of digital tools currently used in your transport network?	>= 3	79	3.86	.888
	< 3	50	3.66	.939

Homogeneous Subsets			
To what extent has digital technology improved transportation efficiency in your supply chain? Tukey HSD ^{a,b}			
What size is your organization?	N	1	2
Large (500+ employees)	75	1.97	
Medium (101-500 employees)	38	2.26	2.26
Small (1-100 employees)	16		2.81
Total	650		217
Means for groups in homogeneous subsets are displayed.			
a. Uses Harmonic Mean Sample Size = 29.359.			
b. The group sizes are unequal. The harmonic mean of the group sizes is used. Type I error levels are not guaranteed.			

Results

- Likewise, the data was thus obtained on the basis of a structured questionnaire sent out via email and LinkedIn to supply chain professionals using purposive sampling to make the information relevant.
- The responses were analyzed quantitatively using Excel by applying descriptive statistics to determine the level of adoption, contribution to sustainability, and barriers.
- Ethical approval, informed consent, and anonymity were followed to the latter to ensure validity, reliability, and generalisability of findings

Has your organization integrated AI or analytics tools to optimize transport routes? - What level of priority does your company place on sustainable transportation practices? - Cross-tabulation									
		What level of priority does your company place on sustainable transportation practices?							
		High	Low	Medium	Very low	Total			
Has your organization integrated AI or analytics tools to optimize transport routes?	Make	Expected Count	7.5	3.6	27.7	33			
	% within row	11.5%	4.8%	73.5%	9.5%	100.0%			
No	Expected Count	2.5	1.2	8.2	1.1	14.0			
	% within row	14.3%	28.6%	42.9%	14.3%	100.0%			
Yes	Expected Count	13.0	6.2	48.1	6.7	73.0			
	% within row	21.5%	6.8%	65.8%	6.5%	100.0%			
Total		Expected Count	23.0	11.0	85.0	110			
		% within row	17.8%	8.6%	65.9%	7.8%			

Tasks	Week 1	Week 2	Week 3	Week 4	Week 5
Selection of the topic and approval proposal					
Primary qualitative data collection and literature review					
Identifying the research methodology					
Collection of data					
Analysis of data					
Findings					
Recommendation and conclusion					
Final submission and celebration					

Multiple Comparisons						
Dependent Variable: To what extent has digital technology improved transportation efficiency in your supply chain?						
Tukey HSD						
(i) What size is your organization?	(j) What size is your organization?	Mean Difference (i-j)	Std. Error	Sig.	95% Confidence Interval	
					Lower Bound	Upper Bound
Large (500+ employees)	Medium (101-500 employees)	-.290	.243	.478	-.86	.30
	Small (1-100 employees)	-.838 ^a	.345	.043	-1.66	-.32
Medium (101-500 employees)	Large (500+ employees)	.290	.243	.478	-.30	.30
	Small (1-100 employees)	-.549	.373	.309	-1.43	.34
Small (1-100 employees)	Large (500+ employees)	.838 ^a	.345	.043	.02	1.65
	Medium (101-500 employees)	.549	.373	.309	-.34	1.43

^a The mean difference is significant at the 0.05 level.

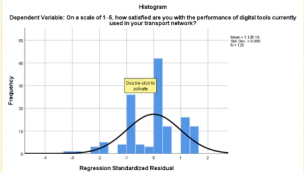
*. The mean difference is significant at the 0.05 level.

Discussion

- Swarm intelligence and multi-agent systems have become effective paradigms in tackling complex coordination and collaboration problems in supply chain management in recent years (Abaku et al. 2024).
- The findings revealed age distinctions, with Gen X being the largest workforce group, whereas both Gen Z and millennials displayed more interest in sustainable mobility, that were representative of the global climate-conscious tendencies.
- The respondents' positions revealed that IT professionals (26.4%) and senior managers (19.4%) play a pivotal role in the digital adoption with the assistance of training and workforce experience.
- The majority of the employees possessed between 4 and 10 years of experience, implying that the moderately experienced employees are vital in technological transformation.
- Organisational size also turned out to be an important variable, with 58.1 percent of the large companies taking advantage of adopting expensive digital solutions over the SMEs.

Model Summary					
Model		Unstandardized Coefficients		Standardized Coefficients	
		B	Std. Error	Beta	1
1	(Constant)	3.425	.363		.000
	To what extent has digital technology improved transportation efficiency in your supply chain?	-.058	.065	-.070	.443
	How familiar are you with the concept of sustainable transportation in supply chain operations?	.134	.134	.089	.318
	How important is real-time tracking in enabling sustainable transport operations?	.076	.069	.101	.273

a. Dependent Variable: On a scale of 1-5, how satisfied are you with the performance of digital tools currently used in your transport network?



Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	12.100 ^a	6	.060
Likelihood Ratio	9.822	6	.132
Linear-by-Linear Association	2.392	1	.127
N of Valid Cases	129		

a. 5 cells (.41 7%) have expected count less than 5. The minimum expected count is 1.08.



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